Earth and Planetary Science Letters, 70 (1984) 383–394
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

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Paleomagnetism of the Xigaze ophiolite and flysch (Yarlung Zangbo suture zone, southern Tibet): latitude and direction of spreading

J.P. Pozzi [1], M. Westphal [2], J. Girardeau [1], J. Besse [1], Yao Xiu Zhou [3], Xian Yao Chen [2] and Li Sheng Xing [3]

[1] *Institut de Physique du Globe de Paris, 4 Place Jussieu, 75230 Paris Cedex 05 (France)*
[2] *Institut de Physique du Globe de Strasbourg, 5 rue Descartes, 67084 Strasbourg Cedex (France)*
[3] *Chinese Academy of Geological Sciences, Beijing (People's Republic of China)*

Received March 16, 1984
Revised version accepted June 29, 1984

The Xigaze ophiolite (29.2° N, 89.5° E), which outcrops in the Yarlung Zangbo suture zone, represents the remnants of an oceanic lithosphere formed in middle Cretaceous times between the Lhasa block to the north and the Indian plate to the south. In an attempt to define the paleo-orientation and latitude of the spreading center at which it has been created, a paleomagnetic study has been done on three sites in volcanics and overlying (or interbedded) radiolarites forming the upper part of the ophiolite sequence and also on seven sites in the Xigaze Group flysch which stratigraphically overlies the volcanics to the north. In each site, hand-blocks carefully oriented both with sun and magnetic compass have been sampled. The paleomagnetism data, combined with structural data on the ophiolite dolerite intrusives, allow a partial reconstruction of South Eurasia at the time of formation of the Xigaze ophiolite. The paleolatitude of accretion and deposition of the Xigaze ophiolite and overlying sediments is found to be 10–20° N. Both ophiolite and basin have encountered a $85 \pm 20°$ anti-clockwise rotation. The corresponding ridge was close to the southern margin of the Lhasa block and was oriented N175 $\pm$ 25°.

1. Introduction

The Yarlung Zangbo river, which runs through South Tibet for several hundred of kilometers in the east-west direction, underlines the main suture zone between Eurasia and India. This suture zone is marked by a sub-continuous belt of ophiolites exemplified by the Xigaze massif (350 km west of Lhasa) which displays a complete ophiolite sequence from mantle peridotites to the south to pillow-lavas stratigraphically covered by deep sediments to the north [1]. A paleomagnetic study of about 100 samples taken in the upper part of the ophiolite sequence and in overlying Xigaze Group flysch sediments is presented. The results are discussed and compared with the paleomagnetic data previously obtained on the Lhasa block. The aim of this study is to define the paleolatitude of accretion of the corresponding ridge and its orientation.

2. Geological setting

The Xigaze ophiolite represents a peculiar oceanic lithosphere comprising from north to south (i.e., from top to base) marine sediments in stratigraphic contact over pillow-lavas or lava-flows to fresh harzburgites and lherzolites [1]. At the top the pillow-lavas are rounded by red manganiferous encrustings giving way first to red radiolarites interbedded with red or green shales, then to white

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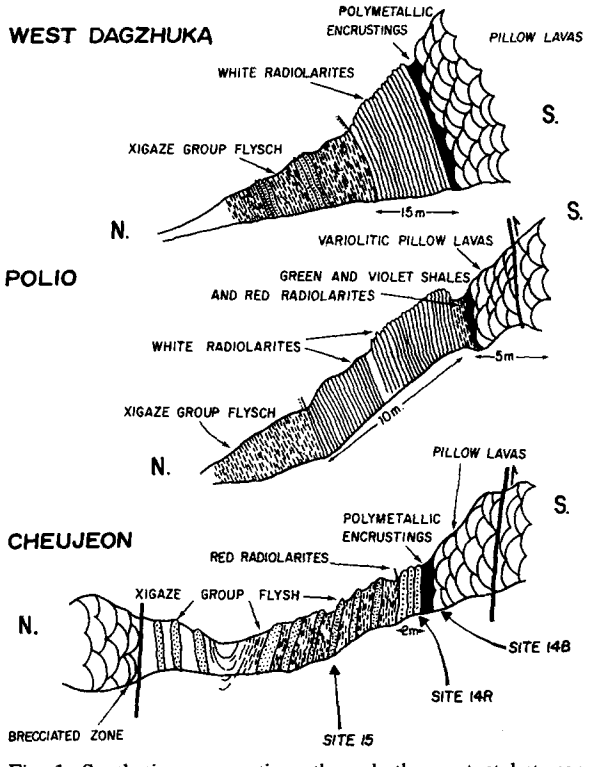


Fig. 1. Synthetic cross-sections through the contact between pillow-lavas, bedded radiolarites and the Xigaze Group flysch (from Girardeau [3]). Only the Cheujeon section has been sampled for paleomagnetism.

well-bedded radiolarites and brown, grey or green volcano-clastic sediments of the Xigaze Group flysch (Fig. 1). The Xigaze Group flysch lies stratigraphically and in conformity on the ophiolites to the south, standing in tectonic contact to the north with a conglomeratic formation (Qiuwu Formation) lying unconformably over the Gangdese plutonic complex [2,3].

Biostratigraphic data on both *Orbitolina* and rudist limestones of the Xigaze Group and on radiolarians from the radiolarite cherts overlying pillow-lavas, give an Upper Albian/Lower Cenomanian age for their deposition [4]. A few south-facing folds can be observed in the Xigaze Group flysch, but the structure of the basin can be interpreted as a large synclinorium with a general fan-like cleavage [5].

Carefully oriented hand-blocks were sampled in twelve sites: a schematic view of the sampling area is given in Fig. 2. Four of them have been sampled in basalts and radiolarites at the top of the ophiolite sequence near Xigaze, Cheujeon and Polio villages. The overlying Xigaze Group sediments have been sampled in five sites located in the region of Xigaze and three near Dagzhuka village. All volcanic samples were oriented both with sun and magnetic compass: many places produced a

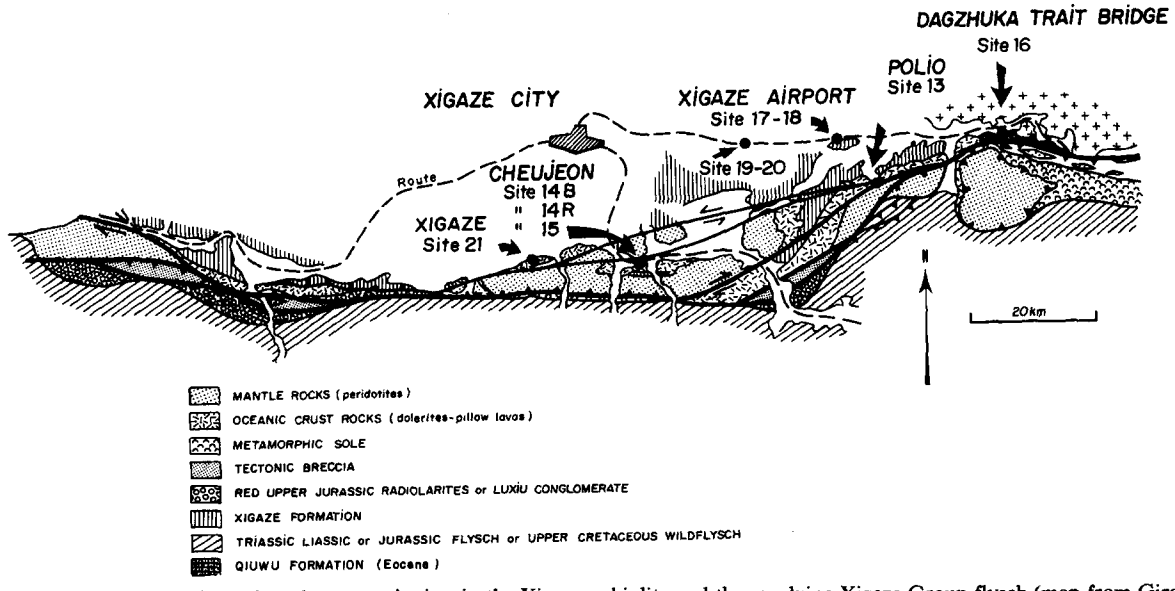


Fig. 2. Location of the main paleomagnetic sites in the Xigaze ophiolite and the overlying Xigaze Group flysch (map from Girardeau [3]).

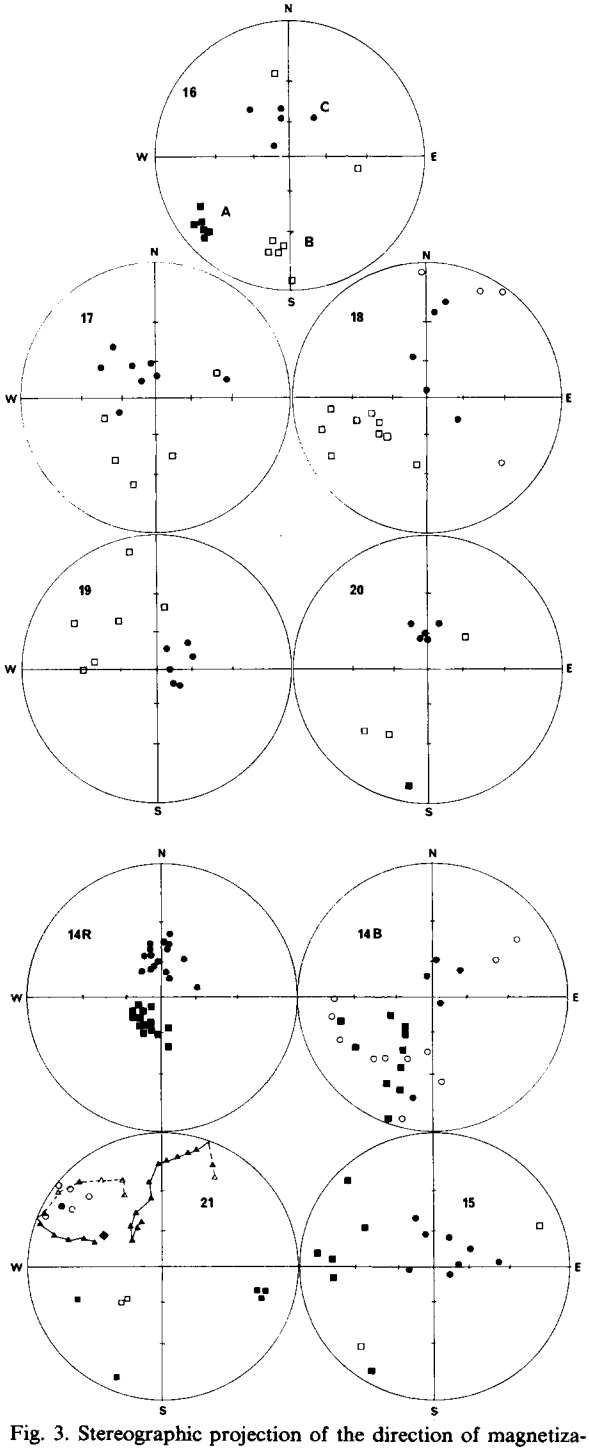


Fig. 3. Stereographic projection of the direction of magnetization (solid and open symbols respectively for positive and negative inclinations; squares and dots respectively for direction of HT magnetization and LT or LF magnetization. An example of demagnetization path is given for site 21; triangles refer to intermediate directions of magnetization. Directions are not corrected for bedding. Diamonds indicate supposed end of demagnetization.

strong deviation of the magnetic compass probably due to lightning and have therefore not been sampled. A site refers to a sampling area over a lateral extension of about 100 m and a vertical one depending on the sampled rocks (20 m in red beds for instance and a few meters in radiolarites). Sites of greater extension have been divided in subsites.

3. Measurements

The samples were drilled in standard 25-mm cores. Volcanics and radiolarites were easily measured with a Digico or a Schoenstedt spinner magnetometer. At least one specimen per sample was demagnetized by stepwise heating and one specimen by alternating field. The different magnetic components were analysed by a least-squares method. A special mention must be made for several flysch samples whose magnetic behaviour was strongly modified during heating: the magnitude of a given magnetization changed, mainly increased due to mineralogical modifications and not due to the demagnetization process.

4. Results

Results are summarized in Table 1 and Fig. 3. All the directions of magnetization are given in Fig. 3 before tectonic correction and in Table 1 after tectonic correction; in most cases the folding axes are nearly horizontal and a classical unfolding algorithm was used. As the fold axis was dipping in the case of sites 14B, 14R and 15 an appropriate tectonic correction was made for these sites. The low blocking temperature components (denoted LT) were usually destroyed between 150 and 350 °C; they were sometimes less scattered than the harder components evidenced above 350 °C and denoted HT (high blocking tempera-

TABLE 1

Xigaze volcanics and sediments (all magnetic components)

Site	Component [a]	Uncorrected		Corrected		N	k	$\alpha_{95}(°)$	
		$D(°)$	$I(°)$	$D'(°)$	$I'(°)$				
13		no characteristic direction							basalts
14B	HT	214	32	239	24	10	13	15	basalts
	LT	221	−19	185	−2	10	5	22	
14R	HT	210	66	279	28	14	37	7	radiolarites
	LT	355	56			16	35	7	
15	HTa	286	15	283	−26	5	14	22	Xigaze Group (flysch)
	HTb	217	−3	219	8	2			
	LT	35	72	341	34	8	8	21	
17	HT	no characteristic direction							Xigaze Group (*Orbitolina* limestones)
	LT	325	58			6	19	16	
18	HT	244	−33	270	23	9	11	15	Xigaze Group (*Orbitolina* limestones)
	LT	18	2	95	30	5	16	24	
19	HT	no characteristic direction							Xigaze Group (flysch)
20	HT	209	−16	229	−52	3			Xigaze Group (flysch)
	LT	353	59			5	73	9	
21	HF	300	36	309	−11	8			basalts
	LF	305	−8	276	−43	6	25	14	
16A	HT	232	12	235	−27	6	189	5	Qiuwu Formation (tuffs—Eocene)
16B	HT	187	−18	170	−59	5	78	9	
16C	HT	no characteristic direction							
16C	LT	348	56			5	13	22	

[a] HT = high-temperature or high-coercivity component; LT = component destroyed at low temperature.

ture or field). It is likely that the hardest component is also older than the softer one.

First we will describe the directions of magnetization found successively for each site and then try to find relations between them.

4.1. *Sites in the upper volcanics of the ophiolite and in immediate overlying sediments (radiolarites and lower parts of the Xigaze Group flysch)*

—*Site 13* consisting of basalts (pillow-lavas) and radiolarites was sampled a few kilometers east of Dagzhuka, near Polio village (Figs. 1, 2). Measurements did not give significant results, all magnetic directions being widely scattered. This site has not been considered.

—*Site 14B*: good demagnetization curves have been obtained from this site. The samples usually show two magnetic components: the first one is destroyed between 150 and 350 °C and the second one only between 550 and 600 °C. Alternating field demagnetization shows very similar results. High blocking temperature components were found in different specimens. Several groups of directions can be isolated for this site (Table 1 and Fig. 3):

(a) A southwest direction with a negative inclination denoted LT, Table 1).

(b) In four specimens, a northward direction with a strong positive inclination.

(c) In two specimens, a direction with northeast declination and negative inclination.

Thermomagnetic curves for site 14B have been set out for several samples. A typical curve is given in Fig. 4 for specimen 14B, first during heating (upper curve), then during cooling (lower curve). The downshift of the cooling curve with respect to the heating curve seems to have a plausible explanation in the decomposition of some component. Titanomaghemite resulting from slight low-temperature oxidation may decompose at the temperature achieved during the experiment. The magnetization after 580 °C seems likely due to paramagnetism. The large irreversible hump on the heating curve between 350 and 400 °C indicates probably a low Curie temperature of 360 °C; a second Curie point appears at 580 °C.

— *Site 14R*: about 20 samples of radiolarites have been taken only a few meters from site 14B in bedded red radiolarites and cherts in perfect continuity with pillow-lavas. They also show two magnetic components. The first one, eliminated below 300 °C, is close to the present dipole field (LT, Table 1). The second component clearly defines a direction near (Table 1 and Fig. 3) $D = 210°$, $I = 66°$, $k = 37$, $\alpha_{95} = 7°$ before tectonic correction.

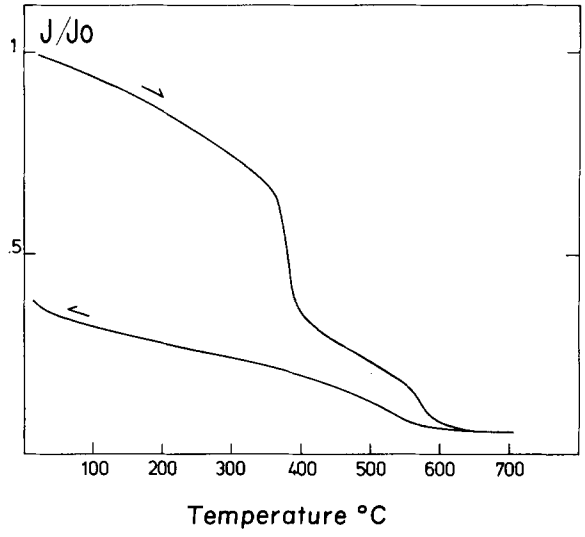


Fig. 4. Thermomagnetic curves for basalt from site 14B (specimen 23). Applied magnetic field: 1.0 T. Upper curve for heating, lower curve for cooling.

The magnetization which corresponds to this direction is cancelled at 650 °C.

Thermomagnetic curves for site 14R: the heating curve has a concave shape below 500 °C suggesting a noticeable contribution of paramagnetism; after a slight hump between 500 and 550 °C the magnetization seems to be due mainly in hematite with a Curie temperature near 670 °C. The cooling curve does not show strong changes produced by heating except for a Curie temperature near 580 °C which is likely a slight contribution of the magnetite produced during heating.

When corrected for tilt, the directions of magnetization found for site 14B and 14R are similar and show positive inclinations of respectively 24° and 28°, both with westward declinations but with a difference of about 40° (Table 1).

—*Site 15* is located a few meters north of site

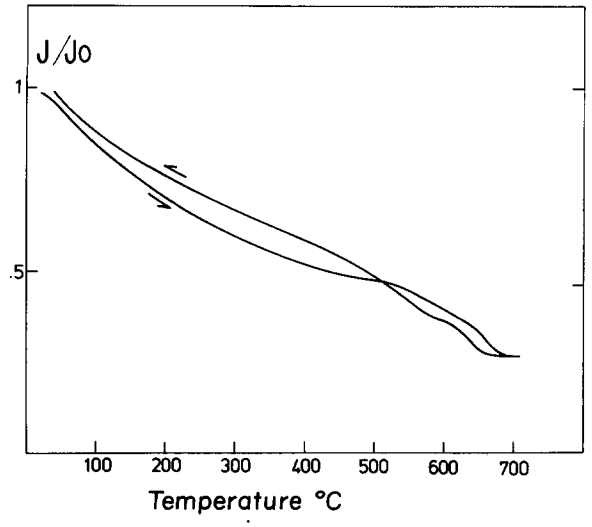


Fig. 5. Thermomagnetic curves for radiolarites from site 14R (specimen 18). Legend as for Fig. 4.

14R (Fig. 1). Sampling has been done in grey and green volcanic sandstones which belong to the lower parts of the Xigaze Group flysch. Sediments of site 15 are in perfect continuity with sediments and volcanics of sites 14R and 14B, respectively. Thermal demagnetization shows two high blocking temperature components. Two samples have a southwest declination with a shallow inclination (HTb) and five samples have a westerly declination with a positive inclination (HTa), before tectonic correction. This latter uncorrected direction, although poorly defined, is close to the corrected direction of site 14R (HT). The thermomagnetic curves of these grey sandstones have varying shapes from specimen to specimen but always show unstable magnetic minerals. We consider that these sandstones may have been remagnetized during or just after folding.

Eight specimens show a LF component of magnetization with a positive inclination and direction comparable to a direction found in site 14B (Fig. 3), not reported Table 1.

—*Site 21* has been sampled in pillow-lavas encrusted by radiolarites (Fig. 2). The samples have been demagnetized by alternating field. The magnetic directions are very scattered but some common features can be found. Six specimens from different samples have a common component destroyed between 30 and 60 mT: this direction is denoted LF (Table 1). Eight specimens show in higher fields (up to 100 mT) a trend towards the direction denoted HF (Table 1): this direction was found by great-circle fitting and intersection. After tectonic correction the HF component is close to the uncorrected LF component (Table 1). This could be interpreted as components acquired during folding.

4.2. *Sites from the northern part of the Xigaze Group flysch close to the Gangdese plutonic complex*

—*Site 16* consisting of andesitic tuffs outcrops east of Dagzhuka, close to the Gangdese plutonic complex (Fig. 2). These tuffs belong to the detritic Qiuwu Formation of probable Eocene age [2,3,5]. It has been divided in three sub-sites being grey (A), grey or pink (B) or pink-coloured (C).

Sub-site 16C did not show any stable high-temperature component while 80% of the natural remanent magnetization was destroyed at 600 °C.

Sub-site 16A gives a southwest direction of magnetization with a shallow positive inclination before tectonic correction.

Sub-site 16B gives a stable direction of magnetization: $D = 187°$, $I = -18°$, not much different from the direction of the LT component for site 14B: $D = 221°$, $I = -19°$, also before tectonic correction. This component is likely a post-Eocene remagnetization.

4.3. *Sites from typical Xigaze Group flysch midway between the ophiolite and the Gangdese plutonic complex*

—*Site 17 and 18* have been sampled near Xigaze Airport in Aptian/Albian *Orbitolina*-bearing limestones (Fig. 2). The intensity of magnetization varies from 2×10^{-3} to 6×10^{-3} A/m. The directions of magnetization of site 17 are rather scattered and poorly defined. LT magnetization lies not far from present field. No other direction can be isolated.

Site 18, different in lithology and much less subjected to water circulation has given better results. The HT component defined by nine blocks

is in a southwest direction (Fig. 3, Table 1). Five specimens give a LT component with a northeast direction, whereas three specimens give strong positive inclinations and have not been considered.

—*Site 19 and 20* have been sampled in fine-grained grey sandstones and in red siltstones respectively, at about 20 km west of Xigaze Airport (Fig. 2). They show a complicated arrangement of directions of magnetization. Site 20 shows a southwest declination and a negative inclination before tectonic correction. Directions of magnetization of site 19 have not been corrected because the tectonic relation between this site and the Xigaze Group flysch has not been clarified; local rotation could have occurred.

5. Discussion

The interpretation of the results is very difficult. The ophiolite and moreover the Xigaze flysch were formed or deposited in an already active zone, folded and thrusted afterwards. It is necessary to decipher which magnetization is of primary origin and which one is the result of a remagnetization produced during or after the folding phase. We nevertheless propose the following interpretation of the results.

The high blocking temperature components (HT) of the Cretaceous sites for which the number of specimens is greater than 5 and for which the α_{95} could be calculated from a first group of directions (component A, Table 2). These directions are given by the HT component of sites 14B, 14R and 18 and give a positive inclination and a westerly declination after correction for bedding. It is difficult to consider a statistical estimation of the fold test with three sites only but it is clear that the fold test is positive. The difference in declination of 40° between sites 14B and 14R could be explained by movements of pillows cooled below Curie temperature. Moreover, the distribution of the HT component directions of site 14B is non-Fischerian indicating that the component HT and LT are difficult to separate. Conversely, the fossilization of the Earth magnetic field by radiolarites and limestones is of better quality.

The low blocking temperature (LT) components of sites 14B and 18, the HTb component of site 15 (Table 1), remagnetized during folding and HT components of Eocene sites 16A and 16B seem to

TABLE 2

	HT component						LT component					
	uncorrected			corrected for bedding			uncorrected			corrected for bedding		
	$D(°)$	$I(°)$	N	$D(°)$	$I(°)$	$\alpha_{95}(°)$	$D(°)$	$I(°)$	N	$D(°)$	$I(°)$	$\alpha_{95}(°)$
Component A												
Site 14B	214	32	10	239	24	15						
Site 14R	210	66	14	279	28	7						
Site 18	244	−33	9	270	23	15						
Component B												
Site 14B							221	−19	10	185	−2	22
Site 18							18	2	5	95	30	24
Site 15	217	−3	2	219	8							
Site 20	209	−16	3	229	−52							
Site 16A (Eocene)	232	12	6	235	−27	5						
Site 16B (Eocene)	187	−18	5	170	−59	9						
Component C												
Site 17							325	58	6			16
Site 20							353	59	5			9
Site 16C							348	56	5			22
Site 14R							355	56	16			7

represent a remagnetization acquired after the main phase of folding (component B, Table 2). The HT component of site 20 can be related to this group.

Before bedding correction, the mean direction of component B is: $D = 211°$, $I = -8°$, $k = 16°$, $\alpha_{95} = 17°$ leading to a paleolatitude of remagnetization $\lambda = 4 \pm 8°$ N acquired after Eocene time.

Finally a group of LT components of magnetization with a north-northwest declination seems to be related to the present dipole field and may be of viscous or chemical recent origin (component C, Table 2).

6. Interpretation

6.1. *Paleolatitude of spreading*

The age of both radiolarites overlying pillow-lavas (sites 14B, 14R) and *Orbitolina*-bearing limestones interbedded in the Xigaze Group flysch (site 18) is middle Cretaceous (Albian/Lower Cenomanian and Aptian/Albian, respectively) [4], just during the long Cretaceous Normal Interval which extends from the Barremian to the Santonian [7]. It is thus unlikely that the directions of magnetization of these formations correspond to a reversed interval of the Earth magnetic field.

For the reasons given in the discussion of component A (Table 2), we determine the paleolatitude of spreading by the direction of magnetization of sites 14R and 18.

The estimated paleolatitude and declination from site 14R is:

$\lambda = 15 \pm 5°$ N $\quad D = 279 \pm 8°$

and from site 18:

$\lambda = 12 \pm 8°$ N $\quad D = 267 \pm 16°$

Statistical treatment is not possible with two directions. We consider that the paleolatitude of accretion and deposition of the Xigaze ophiolite and basin lies between 10° and 20° N.

The rotation encountered by these formations is approximated by: $85 \pm 20°$ anti-clockwise.

6.2. *Paleodirection of spreading*

The direction of the paleospreading center from which the Xigaze ophiolite has been created can be inferred from the orientation of the dikes of sheeted dike ophiolite complexes [8–10]. It is generally accepted that the dikes have been injected in open fissures parallel to the ridges axe as observed in emerged segments of the ridges such as in Iceland [11] or the Asal rift in Afar [12], or in rift valleys of submerged ridges [13,14].

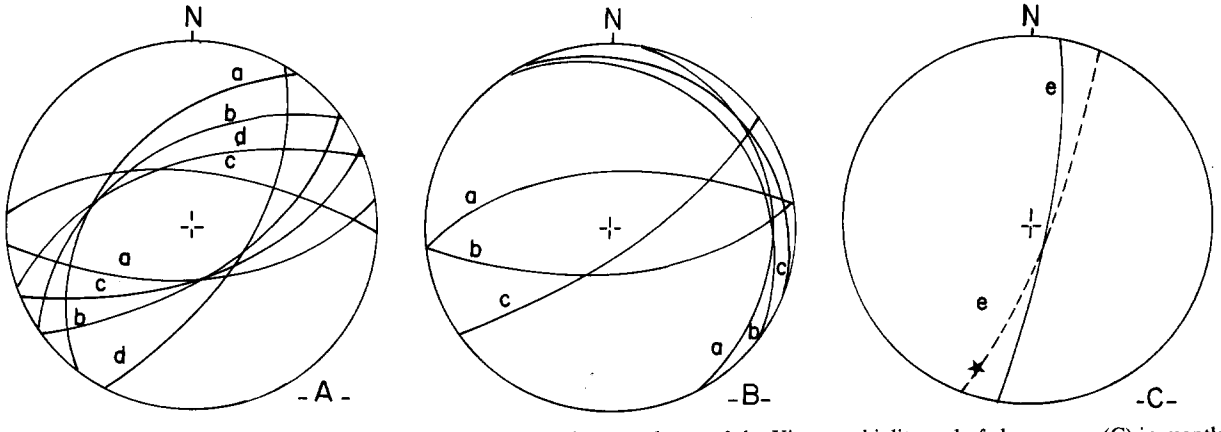


Fig. 6. Attitude of dolerites sills and dikes (A and B) within the crustal part of the Xigaze ophiolite and of shear zones (C) in mantle peridotites (from Girardeau et al. [15]). *a* = Xigaze massif; *b* = Polio massif; *c* = West Dagzhuka massif; *d* = East Dagzhuka massif; *e* = Liugu massif. A: uncorrected for all the massifs (*a* to *e*). B,C: after tectonic correction for *a*, *b* and *c* (for *d* and *e*, no paleohorizontal reference); C: solid line = uncorrected, dashed line = after tectonic correction, star = stretching lineation. Solid lines also represent the computed average planes of sills, dikes or shear zones from different massifs in the Yarlung Zangbo suture zone; equal-area projection, lower hemisphere.

The crustal part of the Xigaze ophiolite mostly consists of dolerite sills locally cross-cutting abundant dolerite dikes [1,14]. The sills and dikes show a constant orientation throughout the whole area studied, over more than 175 km in the east-west direction (Fig. 6A). After tectonic correction of the radiolarites, which have been deposited horizontally and conformably overlie the pillow-lavas, the dolerite intrusives clearly form an horizontal sill complex and a near-vertical dike complex showing an average strike of 080° N with a strong dip to the northwest or southwest (Fig. 6B) [15].

We can therefore consider that the direction of paleospreading center (deduced from field observations) was about N080°, the direction of spreading being about N170°, considering that the direction of spreading is roughly sub-perpendicular to the ridge trend. These present-day orientations combined with the 85° anti-clockwise rotation lead to a paleospreading center oriented N175°, a direction of spreading of N085°. This N085° spreading direction is at a high angle to the flow directions determined within the peridotites which are oriented, after correction, at N130° and N095° in the Xigaze and West Dagzhuka massifs, respectively. But it is actually clear from structural studies of ophiolites that the flow direction beneath a ridge is often strongly oblique to the spreading direction as found by the orientation of sheeted dikes [16].

Some shear zones found within the ultramafic part of the ophiolite have been interpreted, from structural data, as incipient thrust zones formed along transform faults segments of ridges [15] as they stand sub-vertical and show a sub-horizontal stretching lineation after tectonic correction (Fig. 6C). Their strike defined after tectonic correction (Fig. 6C), combined with the anti-clockwise rotation of 85°, gives N110° paleostrikes for these transform fault segments of ridges and these directions are in agreement with the N175° direction of the paleospreading center defined here. The shear sense along the transform faults would be senestral as deduced from the preferred orientations of olivine and orthopyroxene. Such movement is compatible with the eastward motion of Africa relative to Eurasia.

6.3. Relation with the Lhasa block and discussion

It has been shown recently that the red sandstones of the Lhasa block, north of the Yarlung Zangbo suture zone, were located between 20° N and 13° N during Upper Cretaceous times [17–19], in any case far from the Indian continent. Since that time, the sum of the rotations encountered by the Lhasa block is about 30° anti-clockwise.

The paleolatitude of accretion (10–20° N) of the oceanic lithosphere exposed in the Xigaze region and its rotation of 85° anti-clockwise indicates an accretion zone situated near the southern margin of the Lhasa block. The rotation of the Xigaze series is in the same sense but of larger amplitude than the rotation of the Lhasa block. It is therefore likely that the major tectonic contact, which limits the Xigaze Group flysch to the north, with the Gangdese plutonic complex [2,3,5] has absorbed the difference.

The Xigaze ophiolite has been dated as Albian/Cenomanian [4]; at that time, Africa drifted east-southeast relative to Eurasia [20]. This differential movement was likely partially absorbed through the shear zone located south of the Xigaze basin and likely produced east-west extension along the southern margin of South Tibet. We propose that the Xigaze ophiolite represents the remnants of this extension phase. The paleospreading direction and the paleolatitude of accretion of the ophiolite as derived from our data, are in good agreement with this interpretation. This scheme is represented in Fig. 7A; between 110 and 100 Ma a block comprising of Madagascar, India, Antarctica and Australia separated from Africa with a clear southern component of drift. This phase corresponds to an enlargement of the ocean between India and Eurasia and spreading occurred between them (dashed-line ridge) in Fig. 7A [20]. We have added in Fig. 7A the Semail ophiolite (Oman), dated as Albian/Cenomanian, for which the paleolatitude of accretion is about 10° N and the paleo-orientation of the ridge approximately north-south as deduced from paleomagnetic data [21] and from structural data on the metamorphic soles which occur beneath the ophiolite pile [22,23].

During the Middle to Upper Cretaceous, Afghanistan and Iran formed one or several

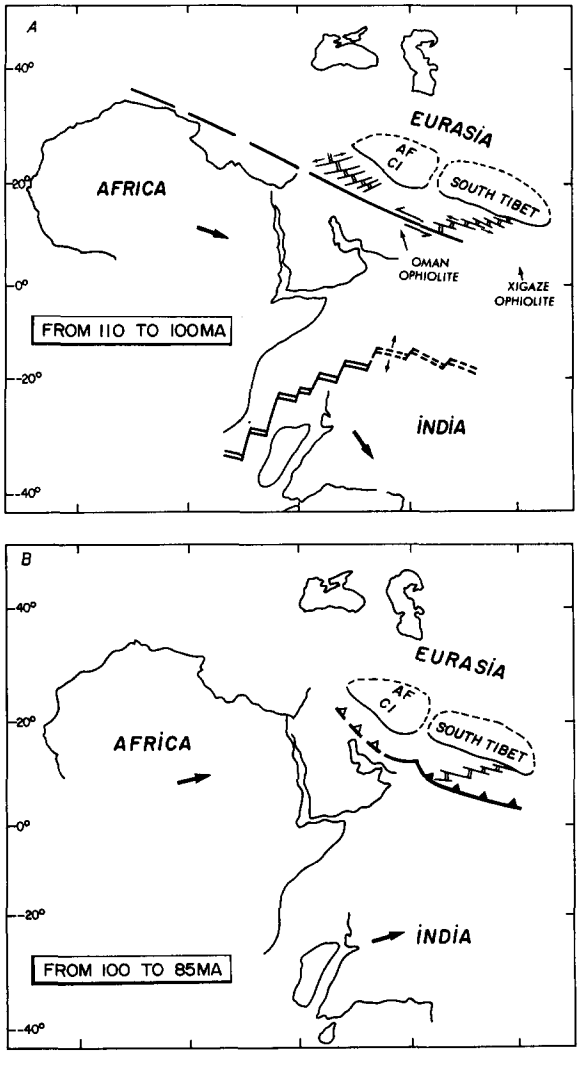


Fig. 7. Tentative reconstruction of the south of the Lhasa block from the Middle Cretaceous to Eocene time. A. Approximately 110 to 100 Ma (AF = Afghanistan; CI = Central Iran). B. 100 to 85 Ma.

juxtaposed blocks along the southern margin of Asia. Their positions can be deduced from paleomagnetic [24,25] and geologic data [26,27]. We have proposed to join South Tibet to Afghanistan on the basis of paleomagnetic results obtained from the Takena Formation [17–19].

The 30° anti-clockwise rotation encountered by the Lhasa block since the Middle Cretaceous has been used to define the northwest-southeast orientation of the southern margin of the Lhasa block. A similar orientation can be deduced from crustal shortening models for southeast Asia and Pamir [28] and from paleomagnetism [29].

Between 100 and 85 Ma (Fig. 7B), India and Madagascar, linked together, did not show relative movement with respect to Africa. Africa and India converged to Europe [20]. For this reason a main subduction zone is proposed along the southern margin of South Tibet, this subduction zone occurring on the main shear zone inferred previously. At that time, the Gangdese plutonic complex commonly ascribed to island arc or continental margin environment and calc-alcalin volcanism just began [30–34]. The Xigaze Group flysch rapidly deposited over the newly created oceanic lithosphere and spreading stopped.

7. Conclusions

The results of this study could be summarized as follows:

—The latitude of accretion of the Xigaze ophiolite and of deposition of its immediate sedimentary cover of Albian/Cenomanian age is between 10 and 20°, close to southern margin of the Lhasa block.

—The accretion of the Xigaze ophiolite is likely related to the nearly west-east motion of Africa relative to Europe, giving rise to an east-west extension near the southern margin of Eurasia.

—The orientation of the paleo-spreading center in which the Xigaze ophiolite was generated was about 175°N and the direction of spreading about 085°N. The spreading center was cross-cut by transform-faults segments oriented at about 110°N.

—They both have undergone an anti-clockwise rotation of about 85 ± 20°. The differences between the two anti-clockwise rotations encountered by the Lhasa block (30°) and the ophiolite and overlying sediments (85°) has probably been absorbed by a major tectonic contact located between the Xigaze Group and the Gangdese plutonic complex.

—The Xigaze ophiolite and flysch have been widely remagnetized after Eocene in a shallow latitude (4 ± 8°N).

The two-stage evolutional model presented here for the creation of the Xigaze ophiolite has just been corroborated by geochemical studies [35].

Acknowledgements

The authors wish to thank J. Marcoux and A. Nicolas for their help in the mapping of the Xigaze ophiolite. This work was supported by the French C.N.R.S. and I.N.A.G. and by the Ministry of Geology of the People's Republic of China during the three-year cooperation project in Tibet. This is a contribution of LA323 and I.P.G.P. (contribution No. 783).

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